Analysis part5

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Linear regression

- When we study the relationship between two (more) variables, other methods than estimating the correlation include studying the effect of a variable over another one (outcome)
- By linear regression the estimate of the effect can be calculated solving the ordinary least squares

Goal: explain a continuous outcome y from predictors (Xi)

Assumptions

- Linearity
- Homoscedasticity (constant variance)
- Normality of residuals
- Independence

Examples

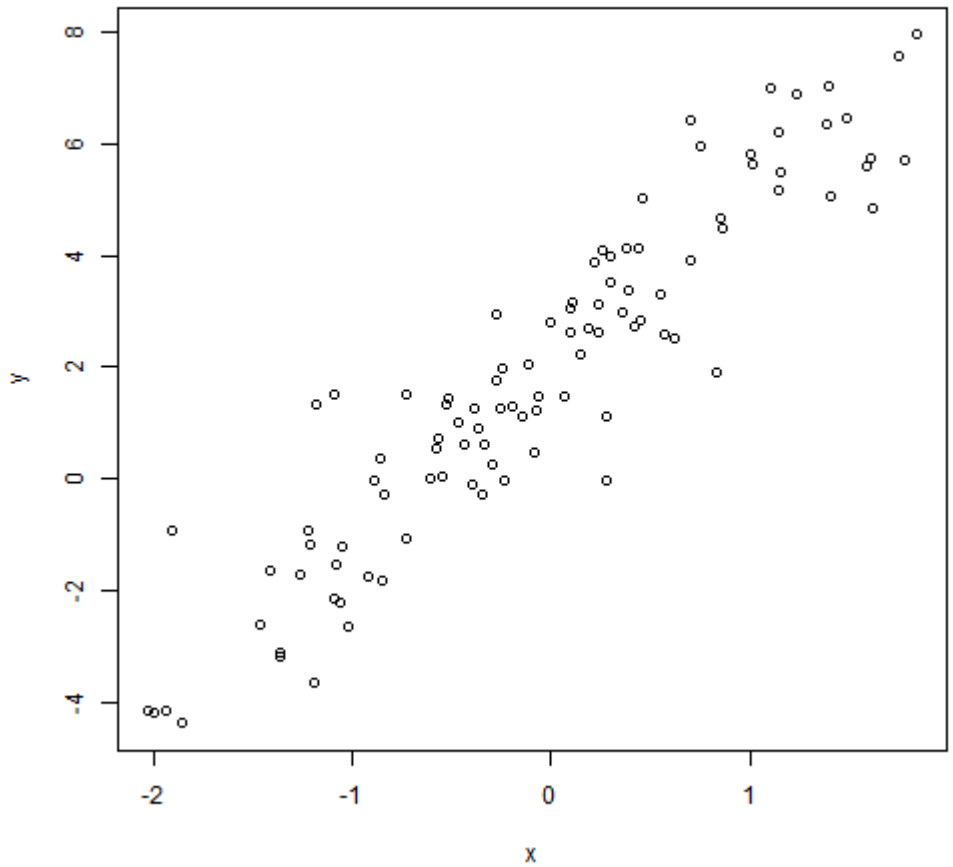
1) We want to investigate how age, weight, and physical activity level affect systolic blood pressure in adults.

- Systolic blood pressure is a continuous variable measured in mmHg.
- We can apply the linear regression model to estimate how much each predictor contributes to variations in blood pressure.

2) We want to study in whether in a hospital there are factors influencing the length of hospital stay after surgery. The outcome variable is the number of days a patient stays in the hospital.

- Predictors include age, type of surgery, and presence of postoperative complications.
- A linear regression model is used to assess how each factor contributes to the duration of hospitalization.

How do we expect a general linear trend?

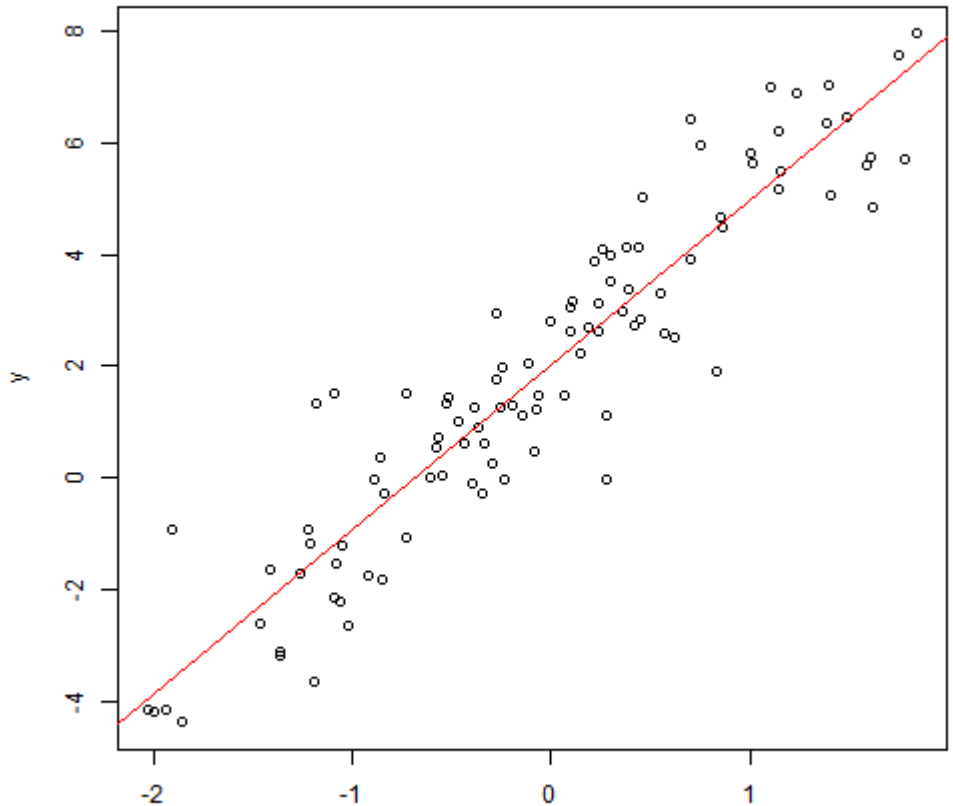


```r
# Fit linear model
lm_model <- lm(y ~ x)
summary(lm_model)
```

```
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.85767 -0.78824 -0.04293  0.65047  2.78808
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.0280     0.1098   18.47   <2e-16 ***
## x             2.9446     0.1158   25.43   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.096 on 98 degrees of freedom
## Multiple R-squared:  0.8684,    Adjusted R-squared:  0.867
## F-statistic: 646.5 on 1 and 98 DF,  p-value: < 2.2e-16
```

Line of trend

```
plot(x, y)
abline(lm_model, col = "red")
```



Logistic regression

Goal: model binary outcome Y

Assumptions

- Independence of observations
- Log-odds is a linear function of predictors

Example

We want to predict the likelihood that a patient has type 2 diabetes based on fasting glucose level, body mass index (BMI), and family history of diabetes. The outcome is binary: presence or absence of diabetes.

- A logistic regression model is used to estimate the probability of diabetes based on these risk factors.

```r
# Fit logistic regression
logit_model <- glm(y ~ x, family = binomial)
summary(logit_model)
```

```
##
## Call:
## glm(formula = y ~ x, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.8730     0.2890    3.02  0.00253 **
## x             2.0262     0.4144    4.89 1.01e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 133.750  on 99  degrees of freedom
## Residual deviance:  89.518  on 98  degrees of freedom
## AIC: 93.518
##
## Number of Fisher Scoring iterations: 5
```

```r
# Predicted probabilities
predicted_probs <- predict(logit_model, type = "response")
```

Logistic regression with classes

1) Used to model class-specific probabilities

- The indicator function: $I(Y=k)$
- I(Y=k) is 1 when the outcome is class k, 0 otherwise.

2) Use when the outcome variable has more than two categories, such as:

- Disease type (e.g., no disease, mild disease, severe disease)
- Risk category (e.g., low, medium, high risk)
- Treatment response (e.g., non-responder, partial responder, full responder)

Types of Logistic Regression:

Multinomial logistic regression: When classes are nominal (no natural order)

Ordinal logistic regression: When classes are ordered (e.g., mild < moderate < severe)

I() is used to force evaluation or treat factors numerically:

```r
# Example: Binary outcome where we model P(Y = 1)
# Using indicator function
glm(I(y == 1) ~ x, family = binomial)
```

```
##
## Call:  glm(formula = I(y == 1) ~ x, family = binomial)
##
## Coefficients:
## (Intercept)            x
##       0.873        2.026
##
## Degrees of Freedom: 99 Total (i.e. Null);  98 Residual
## Null Deviance:        133.7
## Residual Deviance: 89.52     AIC: 93.52
```

```r
set.seed(013)
x <- rnorm(150)
y <- factor(sample(c("A", "B", "C"), 150, replace = TRUE))

# Model one-vs-rest using I()
model_B <- glm(I(y == "B") ~ x, family = binomial)
summary(model_B)
```

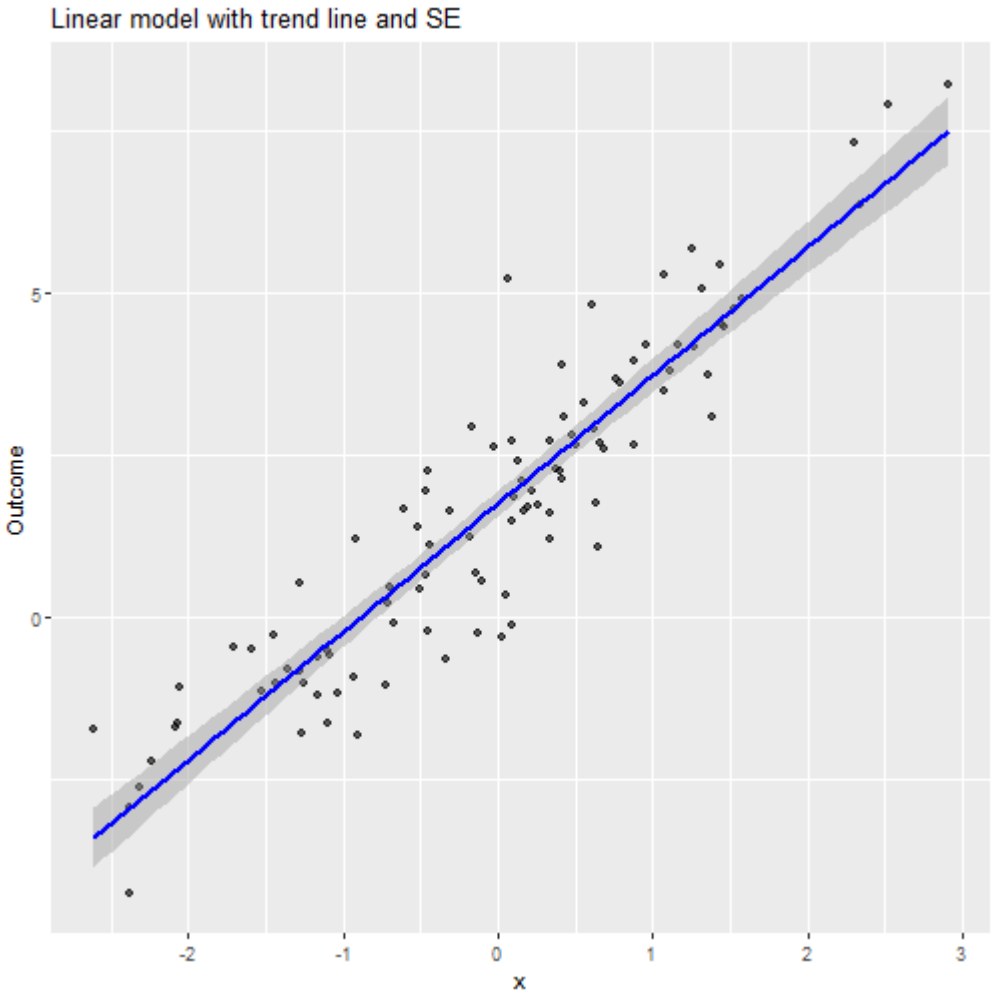
```
##
## Call:
## glm(formula = I(y == "B") ~ x, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.6276     0.1731  -3.627 0.000287 ***
## x             0.2916     0.1849   1.577 0.114806
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 193.61  on 149  degrees of freedom
## Residual deviance: 191.07  on 148  degrees of freedom
## AIC: 195.07
##
## Number of Fisher Scoring iterations: 4
```

Linear and logistic models fit

Model Type Outcome Type Link Function R Function

- Linear Model: Continuous (Identity): **lm()**
- Logistic Model: Binary (Logit): **glm(..., binomial)**
- Logistic + I(): Binary/Multiclass (one-vs-rest Logit) **glm(..., binomial)**

Linear regression model



Interpretation

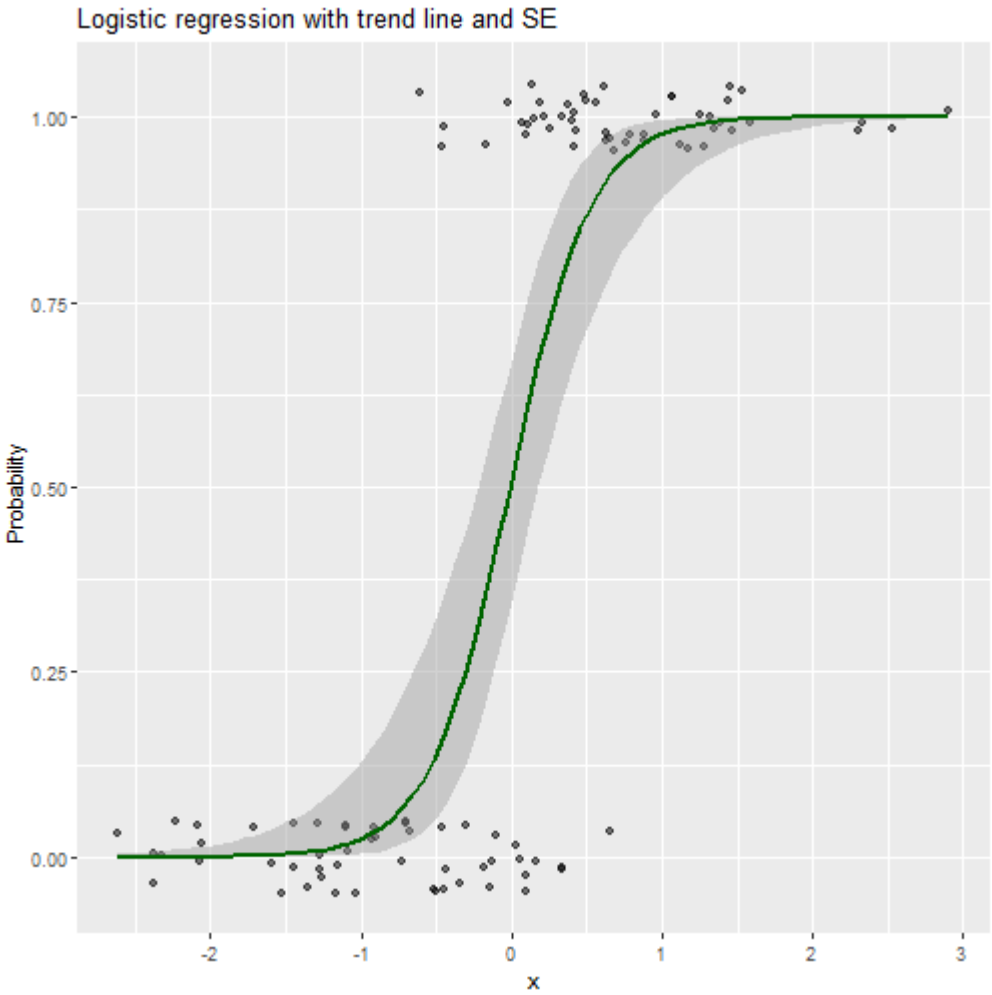
- The **trend line** in a linear regression plot represents the best-fitting straight line through the data points that **minimizes the distance** (error) between the observed values and the predicted values.

Interpretation:

- It shows the expected value of the outcome (Y) for a given value of the predictor (X).
- The slope of the line tells you the direction and strength of the relationship:
- A positive slope: Y increases as X increases.
- A negative slope: Y decreases as X increases.
- The line assumes a constant rate of change (linear relationship) across all values of X.

Meaning of the estimate: the estimated effect of each regressor is the increase (decrease) of the outcome at each unit increase of the regressor. If other regressor are present in the model the effect consider they are kept constant.

Logistic regression model



Interpretation

The **logistic curve** represents the **predicted probability** of an event (e.g., having a disease) as a function of the predictor.

Interpretation:

- The curve is S-shaped (sigmoidal) and ranges between 0 and 1.
- It shows how the probability of the outcome (e.g., probability of having a disease) changes with the predictor.
- It reflects a non-linear relationship between X and the probability:
- At low and high values of X, the probability flattens out.
- Around the middle (inflection point), the probability changes most rapidly.

Meaning of the estimate: the estimated effect of each regressor is the log(OR) of the outcome at each unit increase of the regressor. If other regressor are present in the model the effect consider they are kept constant.